DA2402 · DATA CURATION AND VISUALISATION

How far ahead can you trust a weather forecast?

Forecast error versus lead time in Kolkata, Visakhapatnam and Thiruvananthapuram

PROJECT RESOURCES

GitHub repository: **Weather Forecast Analysis repository**

Data files: **Project data folder**

Interactive dashboard: **Weather Forecast Analysis dashboard**

PROJECT TEAM

Hari Kaeilash · Pardha Saradhi

Collection date: **5 November 2026**

Temperature period: **1 June 2022 – 31 August 2025**

Rainfall period: **20 January 2024 – 31 August 2025**

Forecast model: **GFS seamless · leads 1–7 days**

Executive Summary

This project evaluates archived GFS forecasts from Open-Meteo's Previous Runs API against daily NOAA GHCN-Daily station observations in three Indian cities. The final matched datasets contain 2,787 temperature observations and 822 rainfall observations, with all seven forecast leads available for each retained date.

Pooled temperature MAE increases from 1.57 °C at lead 1 to 1.87 °C at lead 7 (+0.30 °C, approximately 19%). At lead 7, 66% of temperature forecasts remain within 2 °C. Rainfall event skill declines more sharply: CSI falls from 50.2% to 38.6%, and probability of detection falls from 68% to 49%.

The project reports trust horizons of 7.0 days for temperature and 0.9 days for rainfall, using thresholds of MAE ≤ 2 °C and CSI ≥ 50%. The combined horizon is therefore 0.9 days, limited by rainfall. The retained samples cover 78% of the temperature calendar and 46% of the rainfall calendar; rainfall observations are biased towards wet days because dry days are frequently unrecorded.

1 Research Question and Study Design

> **Project question:** How fast does temperature and rainfall forecast error grow with lead time in Indian cities, and is the monsoon harder?

The analysis examines daily maximum-temperature error from one to seven days ahead, rainfall-event verification over the same leads and differences between June–September monsoon months and other months.

Forecast lead is the number of days between forecast issue and valid date. Signed temperature error is forecast minus observed TMAX; MAE is the mean absolute error. A rainfall event is defined as at least 1 mm/day. CSI = hits/(hits + misses + false alarms); POD = hits/(hits + misses); success ratio = hits/(hits + false alarms).

City	NOAA station ID	Station coordinates	Grid distance
Kolkata	IN024141500	22.650° N, 88.450° E	3.4 km
Visakhapatnam	IN001200200	17.717° N, 83.233° E	4.7 km
Thiruvananthapuram	IN010100700	8.467° N, 76.950° E	5.5 km

2 Data Sources and Collection

NOAA GHCN-Daily. Daily station observations were downloaded from the NOAA NCEI by-station directory. The project collected the station list, inventory and three compressed station files. The selected elements are TMAX and PRCP. NOAA GHCN-Daily is identified in the Data.gov catalogue with a CC0 1.0 public-domain dedication; attribution is retained and no NOAA endorsement is implied.

Open-Meteo Previous Runs API. Archived forecasts were retrieved from the Previous Runs API, using the GFS seamless model, station coordinates and Asia/Kolkata timezone. Twelve main requests cover three cities across 2022–2025. Variables are temperature_2m_previous_day1–7 and precipitation_previous_day1–7. Open-Meteo API data are attributed under CC BY 4.0.

API comparison. Six additional checks queried the Historical Forecast and Archive APIs for 2024. Their hourly temperature values matched in only 2.2% of Kolkata hours, 3.2% of Visakhapatnam hours and 2.5% of Thiruvananthapuram hours; mean absolute differences were 1.50 °C, 1.19 °C and 1.40 °C respectively. These endpoints do not provide the required previous-day lead structure, so Previous Runs was used.

The collection contains 27 raw files: NOAA metadata and station files, twelve Previous Runs responses, six API-check responses and robots.txt snapshots. Requests were made through Python notebooks with a two-second delay, response caching and retry handling for HTTP 429 responses.

3 Access Rules, Raw Data and Storage

The saved NOAA robots.txt rules disallow paths beginning /data and /orders; the station files were downloaded from /pub/data/ghcn/daily/by_station/. Open-Meteo's saved robots.txt response is User-agent: * and Disallow: /, which disallows all paths under that host. The project proceeded on the basis of Open-Meteo's free non-commercial API terms, making 18 calls with delays and caching. API terms and robots.txt are separate: the terms do not automatically override the robots.txt restriction. This remains a documented collection-governance consideration.

Raw source	Raw structure	Meaning
NOAA station files	137,187 rows × 8 columns	combined id, date, element, value (tenths), m_flag, q_flag, s_flag, obs_time
NOAA station list + inventory	2 text files	Station identifiers, coordinates and element coverage
Open-Meteo Previous Runs	105,192 hourly rows × 17 columns	city, model, time, seven temperature leads and seven rainfall leads
Historical Forecast / Archive	6 responses	2024 hourly temperature source comparison

The raw NOAA table includes PRCP (54,602 rows), TAVG (38,454), TMAX (24,222) and TMIN (19,909). The raw Open-Meteo table has city, model and time fields plus fourteen forecast variables. NOAA's raw numeric value is stored in tenths for the selected elements; conversion is performed during cleaning.

3.1 Storage structure

Folder	Contents	Purpose
raw/	27 source files, unchanged	Preserve original downloads
interim/	Parsed and type-normalised tables	Intermediate Parquet datasets
clean/	Final temperature/rain tables and data dictionary	Analysis-ready datasets and metadata

Parquet was selected for compact storage and type preservation. The project reports NOAA storage falling from 6.80 MB in CSV to 0.75 MB in Parquet, and Open-Meteo from 10.46 MB to 1.38 MB. Read-back checks confirm identical shapes and dtypes. Final temperature data are 2,787 × 12 (about 49 KB); rainfall data are 822 × 13 (about 25 KB).

4 Data Cleaning and Transformation

The cleaning pipeline parses NOAA YYYYMMDD dates and Open-Meteo timestamps, standardises city and element strings, converts NOAA values from tenths to °C or mm, checks the −9999 sentinel and quality flags, filters the project windows, removes duplicates, rebuilds complete city-date calendars, aggregates forecast hours into daily values and aligns all seven leads. The checks report zero date/time parse failures, zero spelling/case changes, zero numeric values stored as text and zero duplicate rows or duplicate keys.

4.1 Cleaning results

Cleaning stage	Temperature	Rainfall
Full calendar	3,564 city-days	1,770 city-days
After station record endpoint	3,542	1,743
After requiring all 7 forecast leads	3,461	1,722
After removing missing station observations	2,788	822
Final after outlier checks	2,787	822
Final table shape	2,787 × 12	822 × 13
Calendar retention	78%	46%

NOAA is filtered to 1 June 2022–31 August 2025; rainfall analysis begins 20 January 2024. A complete city-date calendar inserts 419 missing city-days as NaN in the temperature calendar. Forecast TMAX is the maximum of 24 hourly values, requiring all 24. Forecast rainfall is summed over the 24 stamps from 09:00 on the previous day through 08:00 on the valid day, to approximate the station's 08:30–08:30 IST reporting window.

4.2 Missing values and treatment

Variable / city	Missing days	Missing rate
TMAX · Kolkata	223 / 1,154	19.3%
TMAX · Thiruvananthapuram	327 / 1,153	28.4%
TMAX · Visakhapatnam	123 / 1,154	10.7%
PRCP · Kolkata	332 / 576	57.6%
PRCP · Thiruvananthapuram	212 / 573	37.0%
PRCP · Visakhapatnam	356 / 573	62.1%

Logistic-regression likelihood-ratio tests reject MCAR for both variables: PRCP p = 8.308e-107 and TMAX p = 7.378e-33. For TMAX, missingness is consistent with MAR conditional on observed month, year and city. For PRCP, missingness may be MNAR because dry days are often not recorded; month and city also contribute. The tests reject MCAR but do not prove MNAR.

No station-value imputation is performed because the station observation is the independent verification reference. Deleting missing station values preserves observed-only evaluation but reduces coverage and may introduce selection bias. The retained rainfall sample is wet-biased: 54% of retained days have at least 1 mm of recorded rain.

4.3 Outlier diagnosis

Temperature outliers are detected using a modified z-score based on the median absolute deviation (MAD) of city-month temperature anomalies, with a threshold of 3.5. Twenty-seven values are flagged; 26 are retained as genuine extremes and one is removed: Kolkata TMAX 21.7 °C on 20 March 2024. Its city-month median is 33.65 °C and neighbouring readings are 32.0 °C and 28.4 °C. A consecutive Visakhapatnam hot spell from 10–17 June 2023 (44.4–44.6 °C) is retained as a real event.

Rainfall outliers are checked using MAD on log(1 + rain) for wet days. The maximum station value is 132.1 mm/day; no value exceeds the 500 mm/day impossibility threshold. No rainfall observations are removed or capped.

5 Forecast Verification Results

5.1 Temperature performance

Pooled temperature MAE rises from 1.57 °C at lead 1 to 1.87 °C at lead 7 (+0.30 °C, approximately 19%). Pooled RMSE rises from 2.05 °C to 2.56 °C. The share of forecasts within 2 °C falls from 72% to 66%. The day-1 to day-7 temperature correlation declines from r = 0.81 to r = 0.71.

City	MAE at lead 1	MAE at lead 7	Change
Kolkata	1.84 °C	2.56 °C	+0.73 °C
Visakhapatnam	1.63 °C	1.73 °C	+0.11 °C
Thiruvananthapuram	1.19 °C	1.26 °C	+0.07 °C

Kolkata contributes most of the increase in temperature error. Its day-7 MAE is 2.56 °C, compared with 1.73 °C in Visakhapatnam and 1.26 °C in Thiruvananthapuram. Kolkata also has the largest day-1 warm bias (+1.00 °C). Grid-to-station separation and local conditions may contribute, but the analysis does not isolate their causal effects.

5.2 Rainfall performance

At the 1 mm/day event threshold, CSI falls from 50.2% at lead 1 to 38.6% at lead 7. POD decreases from 68% to 49%, while success ratio remains around 65–66%. This pattern indicates that longer-lead forecasts miss a greater share of observed wet events. At lead 7, forecast rainfall's 95th percentile is 9 mm compared with 46 mm in the station observations, showing under-representation of heavy rainfall in the retained sample.

6 Monsoon and Seasonal Analysis

Monsoon is defined as June–September. The dashboard compares temperature MAE and rainfall CSI during monsoon months with other months. The reported pooled temperature within-2 °C pass rate is 65% during monsoon and 68% outside it. The monsoon temperature MAE series is lower at the shorter leads and approaches the non-monsoon series by lead 7. Rainfall CSI is higher in monsoon months at lead 1 (53.3% versus 45.6%) and is similar by lead 7 (39.1% versus 37.8%). Therefore, the supplied results do not support a simple claim that the monsoon is uniformly harder for both variables.

A 400-resample bootstrap estimates the monsoon-minus-other-month MAE gap; five of seven valid lead intervals exclude zero. The bootstrap treats days as independent, although weather errors are serially dependent, so the intervals may be optimistic. A block bootstrap by date or month would better account for temporal dependence.

7 Interactive Dashboard

The Plotly dashboard contains 34 plot cards across six tabs: Pulse (5), Temperature (6), Rain (6), Monsoon (6), Cities (5) and Data Quality (6). The dashboard includes error distributions, forecast-versus-observation comparisons, within-tolerance rates, city-by-lead comparisons, rainfall confusion matrices and event metrics, seasonal plots, coverage diagnostics and before/after cleaning comparisons. It uses a consistent dark theme, city colours, aligned cards, tab navigation and explanatory captions.

The dashboard provides more than the rubric minimum of five plots, including univariate and bivariate views and a before/after cleaning comparison. The design uses proximity to group related plots, alignment across panels and a highlight colour to direct attention. Axis labels, units, legends and greyscale readability are part of the dashboard's visual design criteria.

8 Limitations

The findings are limited by non-random missing station observations, particularly rainfall; the 54% wet-day share in the retained rain sample; one NOAA station per city; 3.4–5.5 km station-to-grid separation; slightly different daily reporting windows for station and forecast rainfall; forecast gaps around late December 2023 to late January 2024; one forecast model and seven forecast leads; and project-defined trust thresholds. Rainfall metrics should be interpreted as performance on the retained matched sample, not as an unbiased estimate for every calendar day.

9 Conclusion

The GFS temperature forecasts remain within the project's 2 °C pooled MAE standard through lead 7: MAE increases from 1.57 °C to 1.87 °C, and 66% of lead-7 forecasts are within 2 °C. Rainfall skill declines more quickly: CSI falls from 50.2% to 38.6%, and POD from 68% to 49%. The dashboard's threshold-based horizons are 7.0 days for temperature and 0.9 days for rainfall. Therefore, the combined answer to the project question is approximately 0.9 days, because rainfall—not temperature—sets the shorter trust horizon.

This conclusion applies to the available matched station sample. In particular, the rainfall sample retains only 46% of its calendar and is biased towards recorded wet days, which limits generalisation to all days.

10 Sources, Software and Contributions

Data sources and licences

- NOAA NCEI, Global Historical Climatology Network-Daily (GHCN-Daily), station data directory: NOAA GHCN-Daily data. Data.gov catalogue identifies CC0 1.0 public-domain dedication. Stations: IN024141500, IN001200200 and IN010100700.
- Open-Meteo Previous Runs API: Open-Meteo documentation. API data under CC BY 4.0; attribution: Weather data by Open-Meteo.com. Terms: Open-Meteo terms.

Software and libraries

Python; NumPy 2.4.4; pandas 2.3.3; requests 2.34.2; statsmodels 0.15.0; PyArrow 26.0.0; Matplotlib 3.10.8; Plotly (dashboard visualisation; exact Python package version not recorded).

AI tools

Claude and ChatGPT — coding assistance and slide preparation.

Team contributions

Team member	Contribution
Pardha Saradhi	Data collection, storage and cleaning
Hari Kaeilash	Dashboard and analysis; slide preparation